

Day 4 – Attempted Tasks API Integration & Environment Limitations

Date: 04/02/2026

Duration: approx 4hr and 40min

1. Objective of Day 4

- Continue the EthioSign project by collecting gestures for numbers 0–20.
- Test camera feed, hand detection, and landmark extraction using MediaPipe 0.10.x Tasks API.
- Prepare the pipeline for later CSV recording and dataset collection.
- Evaluate environment compatibility and decide the best approach moving forward.

2. What We Did / Attempted

1. Verified that the **hand_landmarker.task** model file exists in the project folder.
2. Attempted to run a Python script opening the camera to detect hand landmarks.
3. Used `detect_for_video()` from the Tasks API and added timestamp argument.
4. Prepared frame conversion from BGR to RGB and flipped the image for mirror view.
5. Attempted landmark extraction and drawing of landmarks on the camera feed.
6. Evaluated version options for continuing:
 - Option A: Downgrade to Python 3.10 and continue using Tasks API with `Image.create_from_array()` (recommended).
 - Option B: Keep Python 3.12 and try alternative methods, like Android-based landmark collection (less ideal).
7. Chose **Option B** to continue tonight without changing Python, acknowledging limitations.

3. What Went Wrong / Specific Errors

- **Detection error:** `_image_ptr` missing because `detect_for_video()` cannot accept a raw NumPy array in Python 3.12.
- **Import error:** `mediapipe.tasks.python.vision.Image` does not exist in this version.
- No landmarks were detected, so dataset collection failed.

Error messages observed:

```
Detection error: 'numpy.ndarray' object has no attribute '_image_ptr'  
ImportError: cannot import name 'Image' from 'mediapipe.tasks.python.vision'
```

Reason for errors:

- Python 3.12 is incompatible with MediaPipe Tasks API that requires `Image` objects.
- Tasks API expects a MediaPipe `Image` instead of a NumPy array, which is not available in this Python version.
- Using NumPy arrays directly triggers `_image_ptr` errors.

4. Lessons Learned

1. Python version is critical for MediaPipe compatibility.
2. Tasks API cannot run on unsupported Python versions even if the `.task` model is correct.
3. Adding timestamps fixes part of the API requirement but does not solve environment mismatches.
4. Planning for version compatibility saves significant debugging time.
5. Sometimes we must make choices between continuing with environment limitations or changing setup for full functionality.

5. Next Steps / Plan for Day 5

- **Option 1 (recommended, long-term):** Downgrade to Python 3.10 and reinstall MediaPipe 0.10.x Tasks API.
 - Use `Image.create_from_array()` for frames.
 - Collect gestures 0–20, save landmarks to CSVs.
 - Prepare the full dataset for ML training.
- **Option 2 (short-term, current choice):** Keep Python 3.12 and attempt alternative methods.
 - Use Android/phone-based MediaPipe to collect landmarks.
 - Transfer CSV data to PC later.
 - Continue building code structure and CSV pipeline for Day 5.
- Document all environment dependencies for reproducibility.